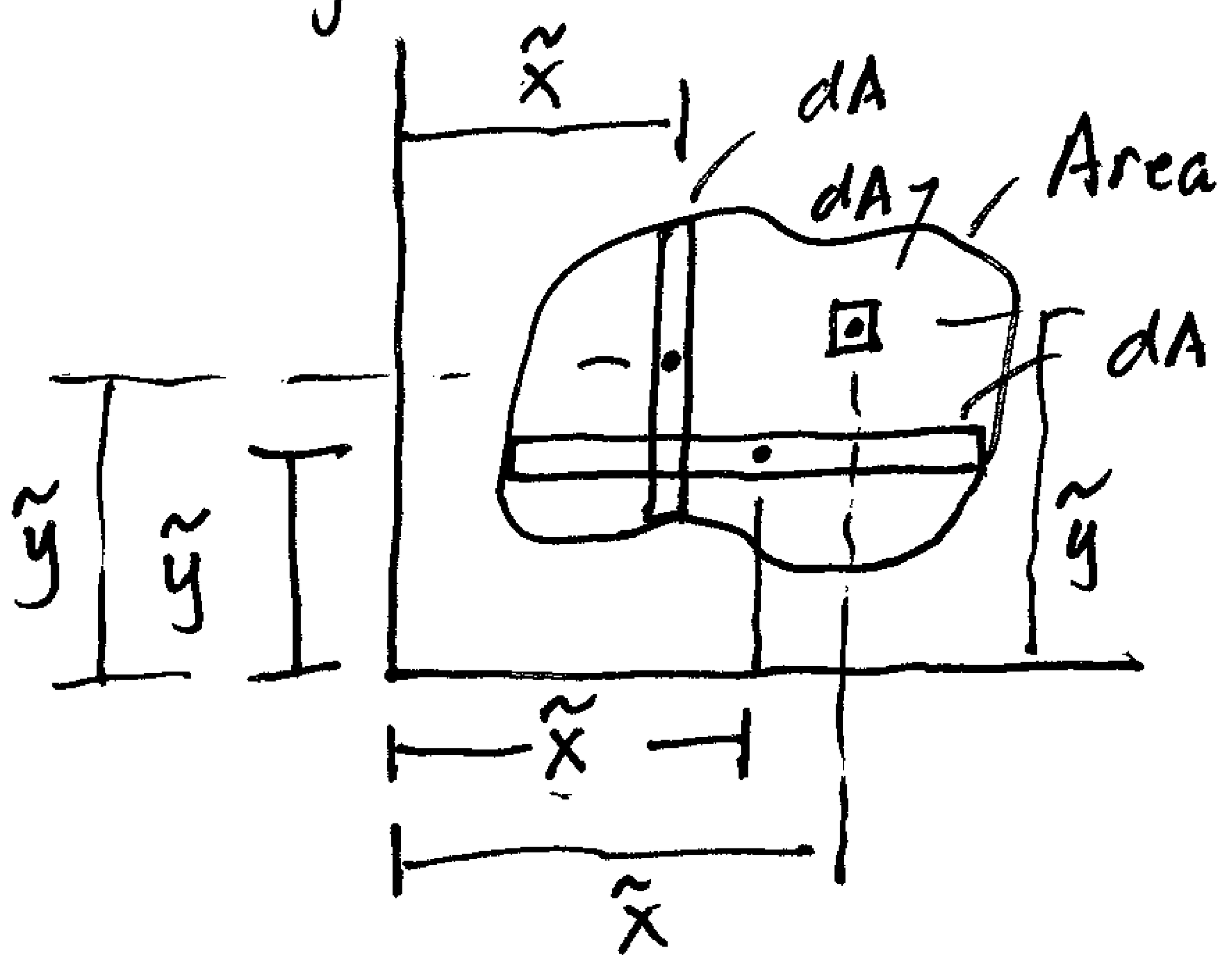
Centroids (Ch. 9)

Integration Formulas:

$$\boxed{\bar{x} = \frac{\int_A \tilde{x}\, dA}{\int_A dA}} \quad \boxed{\bar{y} = \frac{\int_A \tilde{y}\, dA}{\int_A dA}}$$



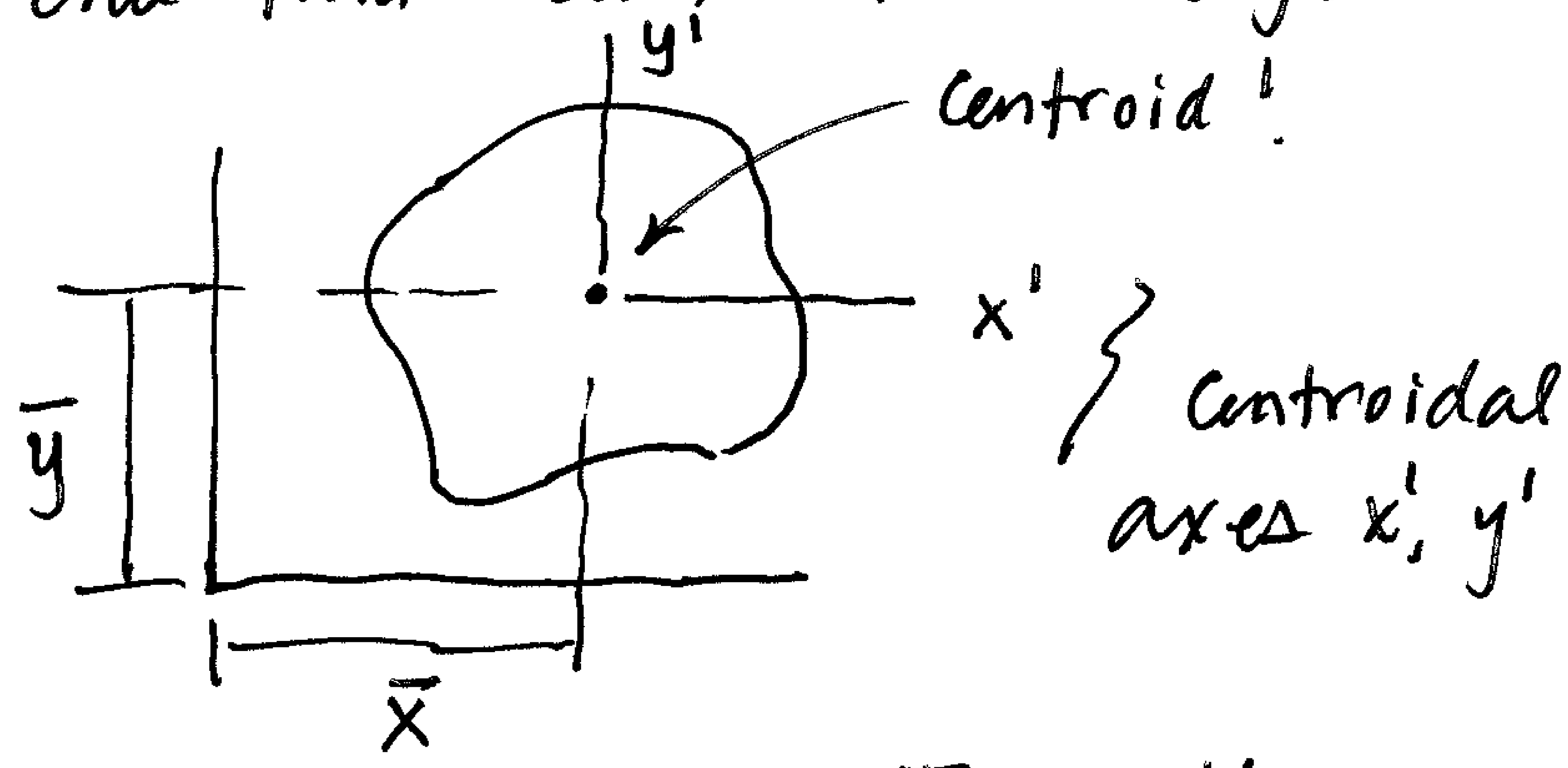
dA can be: ▭, ▯, or ⊡

$\tilde{x}$ = dist from y axis to centroid of dA

$\tilde{y}$ = dist from x axis to centroid of dA . .

__Numerator__ $\int \tilde{x}\, dA$ or $\int \tilde{y}\, dA$ is a __moment equation__!

Once find centroid $(\bar{x}, \bar{y})$:



Then the moments of this area about the x' or y' axes are zero,

i.e. Numerator: $\int_A \tilde{x}\, dA$ about the y' axis = 0.

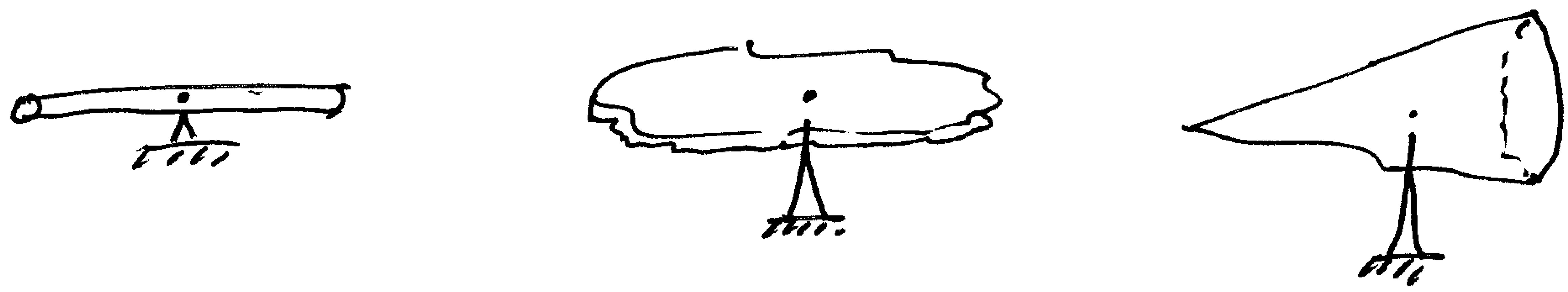
And, $\int_A \tilde{y}\, dA$ about the x' axis = 0.

In other words, the moment of the area on one side of a centroidal axis exactly equals the moment of the area on the other side of the centroidal axis.

In other words...

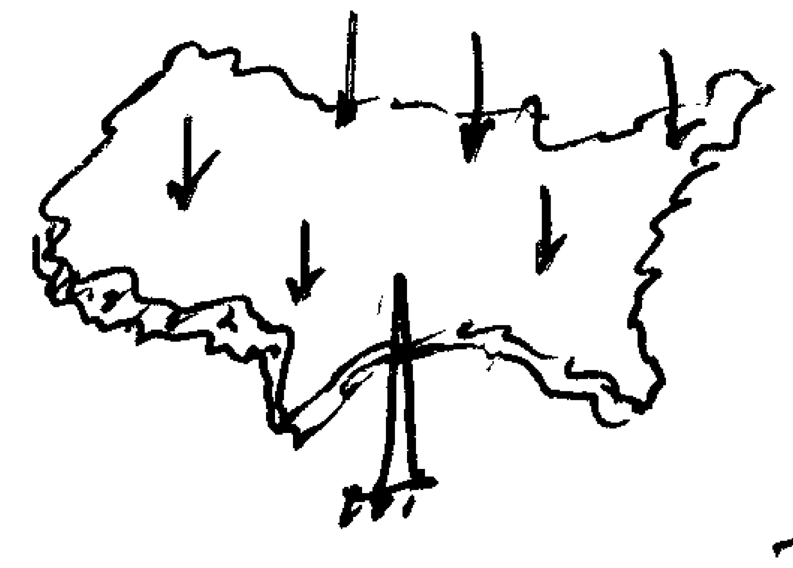
A plate w/ uniform mass per area...
<u>or</u> A rod w/ " " " length...
<u>or</u> A 3-D shape w/ uniform density...

will balance about its centroid.



<u>Examples:</u>

Population Center of USA:

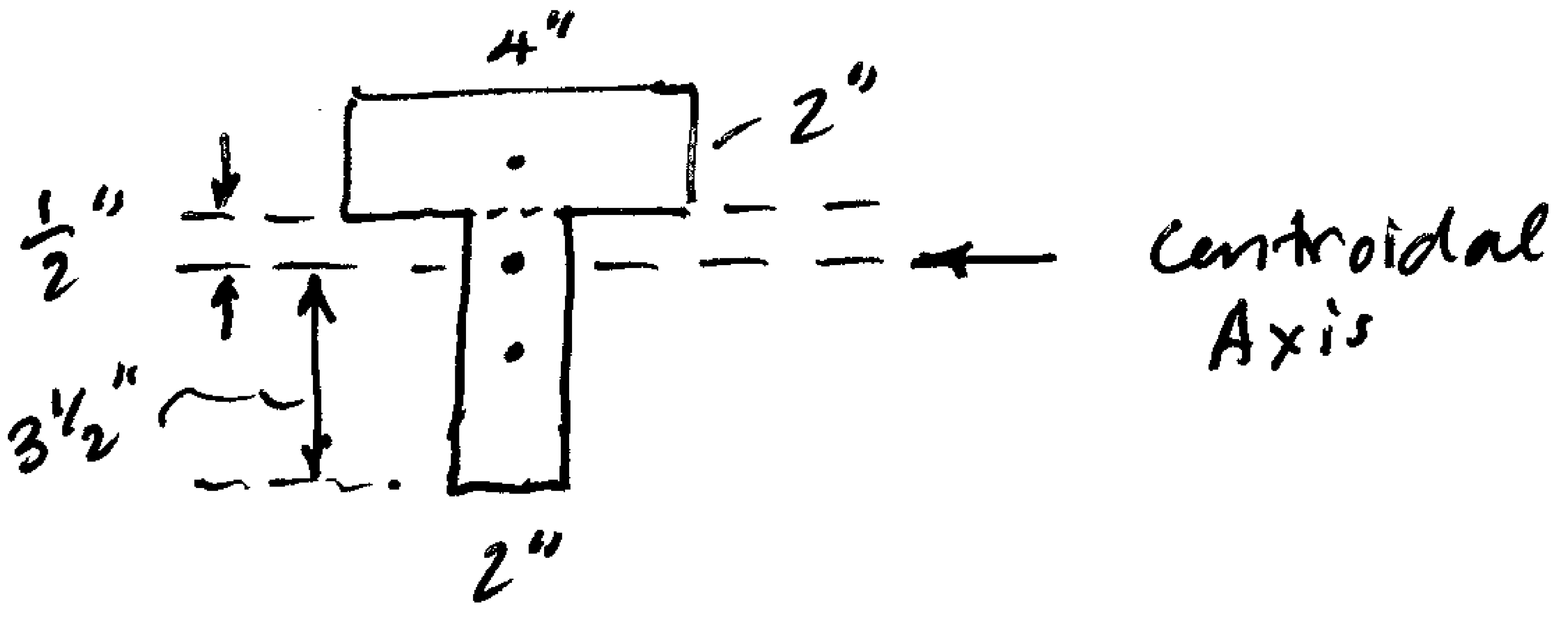


- Each person is a discrete point.

$$\boxed{\bar{x} = \frac{\sum \tilde{x} \cdot 1}{\sum 1}}$$

- The population center of the USA is currently in Southern Missouri!
  Near Hartville, MO:
  ≈ 30 mi E of Marshfield, MO
  42 mi S of Lebanon, MO
  85 mi SW of Rolla, MO

<u>Beam's Centroid</u>

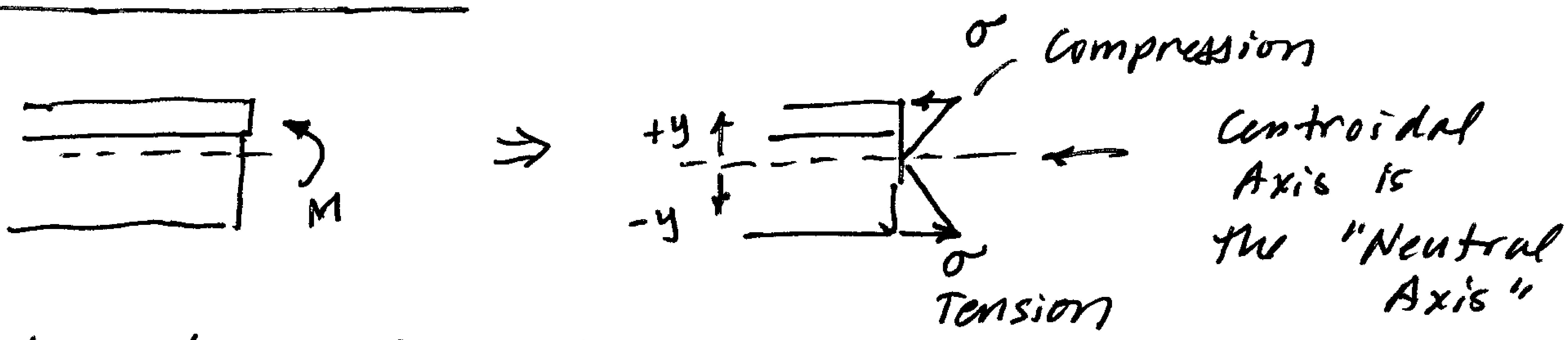


Moment of Area <u>above</u> axis = Moment of Area <u>below</u> axis

$$8(1.5) + (\tfrac{1}{2})(2)(\tfrac{1}{4}) = (3.5)(2)(1.75)$$

$$12.25 \text{ in}^3 = 12.25 \text{ in}^2$$

Mechanics of Materials



Bending stresses increase linearly from neutral axis:

$$\boxed{\sigma = \frac{-My}{I}}$$

Bending stresses (σ) = 0 at Neutral Axis (at centroid!)

Dynamics

Bodies rotate and move about their mass center.

Examples: BMX rider,
Ski Jumper
Spacecraft or satellite...
